Two-transitive coset actions in dimension two

Human Compression Project

Abstract

We derive the two-transitive actions needed in the two fixed Part II originals from the classical subgroup classification of the linear groups of dimension two. A point stabilizer of order b has degree at most $b+1$. This bound, together with a short divisibility argument, removes the subfield alternatives and leaves a small list of exceptional actions. The resulting common argument also determines the possible overgroups of the small intersections used in those originals. We do not assume their Proposition 4.6 or its unpublished reference, and we do not count embeddings or pairs of parabolics here.

1 Premises and the elementary bound

We write D_{2n} for a dihedral group of order $2n$. The classical subgroup theorems of Dickson, Moore and Suzuki are borrowed premises: the proper subgroups of $\mathrm{PSL}(2,t)$ are cyclic or dihedral groups, elementary abelian groups and their scalar cyclic extensions, A_4, S_4, A_5, and the subfield groups $\mathrm{PSL}(2,s)$ and $\mathrm{PGL}(2,s)$. We use their usual field restrictions, maximal-subgroup lists and normalizers. In particular a subgroup $\mathrm{PGL}(2,s)$ in $\mathrm{PSL}(2,t)$, for odd s, requires an even extension degree. The normalizer of a four-group in odd characteristic contains an A_4. A torus has its dihedral normalizer. The two S_4 classes in $\mathrm{PSL}(2,7)$ and the two A_5 classes in $\mathrm{PSL}(2,9)$ and $\mathrm{PSL}(2,11)$ are interchanged by $\mathrm{PGL}(2,t)$. These are subgroup and fusion premises, not assertions about the geometries to be recovered. They are identified separately in Sections 4.1, 4.2 and 4.4 of the fixed published source [1]. We retain that source's attribution to [2, 3, 4]; we do not represent the historical books as newly checked source editions. The modern statement in [5, Theorem 2 and Lemmas 1–2], consulted separately, also records the subgroup types in odd characteristic and the small exceptional subgroup lattices. Its entire design theory is not an input to this proof.

We also use the classical isomorphisms $\mathrm{PSL}(2,4) \cong \mathrm{PSL}(2,5) \cong A_5$, $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$, $\mathrm{PSL}(2,9) \cong A_6$ and $\mathrm{PGL}(2,5) \cong S_5$. The simplicity of $\mathrm{PSL}(2,t)$ for $t \geq 4$ is retained as a premise. The small subgroup lattices of A_4, S_4, A_5 also follow directly from their natural permutation actions; we use their maximal subgroup lists below.

Lemma 1. *Suppose that H acts two-transitively on the cosets of a proper subgroup B, and put $d = [H : B]$. Then B is maximal in H, $d-1$ divides $|B|$, and $|H| \leq |B|(|B|+1)$.*

Proof. A nontrivial system of blocks would prevent a point stabilizer from being transitive on the other points. Thus the action is primitive, and B is maximal. The B-orbit on the other $d-1$ points has length $d-1$, which divides $|B|$. Multiplying $d \leq |B|+1$ by $|B|$ proves the bound. Faithfulness is unnecessary. □

2 The simple groups

Theorem 2. *Let $H = \mathrm{PSL}(2,t)$, where $t \geq 4$ is a prime power. The two-transitive coset actions are the natural action with point stabilizer $E_t : C_{(t-1)/(2,t-1)}$, and the following exceptional actions, with repetitions under the small isomorphisms suppressed:*

H	B	$[H : B]$
A_5	A_4	5
A_5	D_{10}	6
$\mathrm{PSL}(2,7)$	S_4	7
$\mathrm{PSL}(2,9)$	A_5	6
$\mathrm{PSL}(2,11)$	A_5	11.

Here each embedding class of the displayed stabilizer gives its corresponding coset action. The assertion is not a count of ambient embeddings into a larger group.

Proof. The kernel is normal in the simple group H and is contained in the proper subgroup B, hence is trivial. Apply the maximal-subgroup list and Lemma 1.

For a torus normalizer the two possible degrees are $t(t-1)/2$ and $t(t+1)/2$. For odd $t \geq 7$, their point stabilizers have order at most $t+1$, whereas $t(t-1)/2 - 1 > t+1$. For even $t \geq 8$, their order is at most $2(t+1)$ and $t(t-1)/2 - 1 > 2(t+1)$. Thus only $t = 4, 5$ remain. In A_5, D_{10} has index six; D_6 has index ten, which fails $9 \mid 6$.

If $B = A_4$, the order bound is $|H| \leq 156$, so $t \leq 6$ in odd characteristic and $t \leq 5$ in even characteristic. Only $t = 4, 5$ give the required subgroup. If $B = S_4$, the bound is $|H| \leq 600$. The only possible fields with an S_4 subgroup are $t = 7, 9$; for $t = 9$ the degree is fifteen and $14 \nmid 24$. If $B = A_5$, the bound is $|H| \leq 3660$. Apart from $H = A_5$, the possible orders divisible by sixty are those for $t = 9, 11, 19$; the last has degree fifty-seven and $56 \nmid 60$. These finite checks use the order $t(t^2-1)/(2, t-1)$ and the prime powers in the stated bounds.

For an even field, a maximal proper subfield group has $t = s^r$ with r prime. At $r = 2$ its index is $s(s^2+1)$ and its order is $s(s^2-1)$, so the index minus one already exceeds its order. The index increases with r. For an odd field and a maximal simple subfield group, $r \geq 3$. Its index is at least $s^2(s^4 + s^2 + 1)$, again exceeding its order plus one. The remaining maximal subfield possibility is $\mathrm{PGL}(2,s)$ in $\mathrm{PSL}(2,s^2)$ for odd $s \geq 3$. Its order is $b = s(s^2-1)$ and its index is $d = s(s^2+1)/2$. But

$$1 < \frac{b}{d-1} = 2 - \frac{4s-4}{s^3+s-2} < 2.$$

Consequently $d - 1$ cannot divide b. This removes every proper subfield alternative.

The listed actions do exist. Translations in a Borel subgroup are transitive on the finite points of the projective line, proving the natural case. The natural A_5 action on five letters is two-transitive. Its action on the six Sylow five-subgroups is two-transitive because a Sylow five-subgroup fixes only itself: another fixed point would put it in a different normalizer of order ten. It therefore has one orbit on the other five points. The action of $\mathrm{GL}(3,2)$ on its seven nonzero vectors is two-transitive, since two distinct such vectors are independent. The natural A_6 action gives the degree-six case.

For $H = \mathrm{PSL}(2,11)$ choose one class of A_5 subgroups. It has eleven members. There are fifty-five A_4 subgroups, all conjugate and self-normalizing; each chosen A_5 contains five. Double counting shows that each A_4 lies in exactly one member of the chosen A_5 class. Hence distinct members do not intersect in an A_4. In the action on that class the stabilizer A_5 has no additional

fixed point, because the normalizer of an A_5 is itself. Its possible nontrivial orbit lengths at most ten are five, six and ten, by the subgroup orders of A_5. Length five would give an A_4 intersection and is impossible. Ten remaining points cannot be partitioned into orbits of length six. They form one orbit of length ten. This proves two-transitivity, for either of the two A_5 classes. □

3 The projective groups

Theorem 3. *Let $H = \mathrm{PGL}(2,t)$ for odd $t \geq 5$. Its two-transitive coset actions have point stabilizer $\mathrm{PSL}(2,t)$ and degree two, the natural Borel $E_t : C_{t-1}$ and degree $t+1$, or, when $t = 5$, S_4 and degree five. For $\mathrm{PGL}(2,3) = S_4$ the stabilizers are A_4, D_8, S_3, of degrees two, three and four.*

Proof. Put $N = \mathrm{PSL}(2,t)$. If N is in the kernel, the nontrivial action factors through $H/N = C_2$, giving the first case. Otherwise a nontrivial kernel would meet the simple group N trivially and centralize it. The centralizer of N in H is trivial: commuting with all translations fixes their unique common fixed point and gives an affine map with scalar one; commuting also with the opposite unipotent subgroup makes its translation zero. Thus the action is faithful.

Primitivity now makes N transitive. For a point stabilizer B write $C = B \cap N$. Then $H = NB$, $|B| = 2|C|$, and $d = [H : B] = [N : C]$. In particular

$$d - 1 \mid 2|C|, \qquad |N| \leq |C|(2|C| + 1). \tag{1}$$

We apply the full subgroup list to C, which need not be maximal in N.

If C has nontrivial unipotent radical $O_p(C)$, this radical is characteristic in C and has a unique common projective fixed point. Since $C \lhd B$, B fixes this point. Maximality of B in H makes B the full Borel, giving the natural action. This covers all affine types, including nontrivial unipotent cyclic subgroups.

If C is a nontrivial semisimple cyclic group, B normalizes its torus. For an involution this follows by centralizing it, which preserves its two eigenlines as an unordered pair. If C is dihedral with rotation group of order greater than two, that rotation group is characteristic, with the same conclusion. Thus maximality forces B to be a torus normalizer. Its order is at most $2(t+1)$ and the smaller possible degree is $t(t-1)/2$. For $t \geq 7$, $d - 1 > 2(t+1)$. At $t = 5$ the two degrees are ten and fifteen, with stabilizer orders twelve and eight, and both fail divisibility. The case $C = 1$ fails (1). If C is a four-group, its normalizer contains an A_4, while $|B| = 8$. This normalizer is proper, since C cannot be normal in the simple group N. It would strictly contain the maximal subgroup B, a contradiction.

If $C = A_4$, (1) gives $|N| \leq 300$. Only $t = 5, 7$ can occur; $t = 7$ gives $d = 14$ and $13 \nmid 24$. For $t = 5$ the extension is S_4 inside S_5, with its natural degree-five action. If $C = S_4$, the bound is $|N| \leq 1176$. The possible fields are $t = 7, 9$; $t = 9$ fails $14 \mid 48$. At $t = 7$, an element of $B \setminus N$ would normalize the S_4 subgroup, contrary to the interchange of its two N-classes by H. If $C = A_5$, the bound is $|N| \leq 7260$. The proper possibilities are $t = 9, 11, 19$; $t = 19$ fails $56 \mid 120$. For $t = 9, 11$, interchange of the two A_5 classes again excludes such a B. The case $C = N = A_5$ belongs to the already separated degree-two action.

Finally suppose C is a proper subfield group with field size s. For $C = \mathrm{PSL}(2,s)$ the smallest possible index occurs at $t = s^2$ and equals $s(s^2+1)$; this already exceeds $2|C|+1 = s(s^2-1)+1$. For $C = \mathrm{PGL}(2,s)$, t is an even power of s. At $t = s^2$,

$$3 < \frac{2|C|}{d-1} = 4 - \frac{8s-8}{s^3+s-2} < 4 \qquad (s \geq 3),$$

so divisibility fails. At any larger allowed extension degree, even the index for $t = s^3$ is $s^2(s^4 + s^2 + 1)/2 > 2s(s^2 - 1) + 1$, and the bound fails. This exhausts C.

The Borel and degree-two actions plainly exist, and the exceptional degree-five action is the natural S_5 action. The maximal subgroups A_4, D_8, S_3 of S_4 give respectively its quotient action on two points, its S_3 quotient on three points, and its natural action on four points. They exhaust the case $t = 3$. □

4 Solvable overgroups

The following elementary observations handle the remaining Dickson types without a classification of general two-transitive groups.

Lemma 4. *For a dihedral group, a two-transitive coset action has degree two or three. In degree two the stabilizer is any subgroup of index two; in degree three it is a dihedral subgroup of index three.*

Let $H = V : C$ be a scalar affine subgroup in characteristic p, where $V \neq 0$ is elementary abelian and C is cyclic of order prime to p. A two-transitive coset action either has degree two and its stabilizer contains V, or has a stabilizer conjugate to $W : C$, with W a C-invariant subgroup of V and

$$[V : W] = |C| + 1.$$

The induced scalar action on $V/W \setminus \{0\}$ is regular.

Proof. For a dihedral group, the orbits of its normal rotation group form blocks. If this group is in the kernel, the action factors through C_2. Otherwise it is transitive in the faithful quotient. A transitive abelian permutation group is regular, since all point stabilizers coincide and hence lie in the kernel. The point stabilizer in the resulting dihedral permutation group therefore has order at most two. Transitivity on the other points implies that the degree is at most three. A subgroup of index three with a transitive rotation group contains a reflection and is dihedral. Conversely the quotients on two and three points are C_2 and S_3, respectively.

For the affine group, primitivity likewise says that V is in the kernel or is transitive in the faithful quotient. In the first case the action factors through the abelian group C; two-transitivity then forces degree two. In the second case put $W = V \cap B$. Regularity of the image of V shows that W is the kernel within V, hence is H-invariant, and $H = VB$. The image of B in $H/V = C$ is all of C. Modulo W, the subgroup B/W is a complement to V/W. Such complements are conjugate by V/W: one may average an affine fixed point over the cyclic complement, since $|C|$ is invertible in characteristic p. Thus B is conjugate to $W : C$.

For a nonidentity scalar c, multiplication by $c - 1$ is invertible on V and maps the invariant finite subspace W onto itself. It is therefore invertible on V/W. Thus the nonidentity elements of C have no nonzero fixed vector in V/W, and two-transitivity says that this free action on the nonzero vectors is also transitive. This proves the formula. □

5 The small intersections in the fixed originals

Corollary 5. *Within the field scopes used in the published Part II source, the possible proper two-transitive overgroups of the indicated intersection are as follows. Each listed dihedral overgroup*

requires its usual torus divisibility condition.

B	*ambient field scope*	H
D_{10}	*the maximal* A_5 *cases*	D_{20}, D_{30}, A_5
A_4	*the maximal* A_5 *cases*	$S_4, A_5, E_{16} : C_3$
D_6	q *prime,* $q \equiv \pm 1 \pmod 8$	D_{12}, D_{18}, S_4
D_8	q *prime,* $q \equiv \pm 1 \pmod 8$	D_{16}, D_{24}, S_4
A_4	q *prime,* $q \equiv \pm 1 \pmod 8$	S_4, A_5
A_5	$q = 9^m$ *or* 11^m, m *odd prime*	$\mathrm{PSL}(2,9)$ *or* $\mathrm{PSL}(2,11)$
S_4	$q = 7^m$, m *odd prime*	$\mathrm{PSL}(2,7)$
D_8	$q = 9$	S_4
S_4	$q = 25$	$\mathrm{PGL}(2,5)$.

The last two alternatives in the A_5 row are restricted to their actual defining characteristic. The affine $E_{16} : C_3$ alternative occurs only in characteristic two and needs at least two dimensions over $\mathbb{F}_4$ in the ambient additive field. An S_4 or A_5 alternative is included only when it embeds in the specified ambient group.

Proof. For the cyclic and dihedral Dickson types, a nonabelian stabilizer cannot lie in a cyclic group. Lemma 4 gives exactly the displayed double and triple dihedral orders. For $q = 9$ neither D_{16} nor D_{24} embeds: the maximal tori have orders four and five.

Consider an affine overgroup. Its stabilizer either contains the whole additive kernel with index two, or is $W : C$ as in Lemma 4. For $B = A_4$, the latter forces $W = E_4$, $C = C_3$, and $[V : W] = 4$, giving $E_{16} : C_3$. An index-two extension would instead require a faithful scalar C_6 on E_4, impossible since its scalar group has order three. This also shows directly that no affine alternative exists in odd characteristic for $B = A_4$.

For $B = D_{10}$, a nontrivial additive normal subgroup must be C_5. The transitive-kernel alternative would demand an additive quotient of size $|C_2|+1 = 3$ in characteristic five, impossible. The index-two alternative would be $C_5 : C_4$. In the present maximal-A_5 scope a characteristic-five field has size 5^m with m odd, so its scalar torus has order $(5^m - 1)/2$, which is twice an odd number. It contains no C_4. For $B = D_6$ the affine possibility would require characteristic three; the stipulated prime $q \equiv \pm 1 \pmod 8$ is not three. For D_8 in odd characteristic there is no nontrivial normal subgroup of odd prime-power order. Finally S_4 and A_5 cannot occur as these affine stabilizers: in odd characteristic they have no nontrivial normal subgroup of defining prime-power order, and they are not cyclic. Thus no affine cases were lost.

The small exceptional overgroups are read directly from the maximal-subgroup lists: A_4 has its degree-four action with stabilizer C_3; S_4 has the three actions in Theorem 3; A_5 has the degree-five and degree-six actions in Theorem 2. These give precisely the S_4 and A_5 entries above.

For all larger subfield overgroups, apply Theorems 2 and 3 to the abstract overgroup. Reading off the displayed stabilizers gives $\mathrm{PSL}(2,7)$ above S_4, $\mathrm{PSL}(2,9)$ and $\mathrm{PSL}(2,11)$ above A_5, and $\mathrm{PGL}(2,5)$ above S_4 or its normal A_5. Such groups are neither cyclic, dihedral, affine nor one of the three small exceptional Dickson types, so their embeddings must be subfield embeddings in the actual defining characteristic. This excludes $\mathrm{PGL}(2,5)$ from the characteristic-three and -eleven A_5 rows, and from the characteristic-seven S_4 row. It excludes $\mathrm{PSL}(2,7)$ in characteristic five. The natural Borel cases give no further small stabilizers. This proves the table. □

6 Source use and limits

Corollary 5 proves the overgroup possibilities used in Lemmas 5.1, 5.2, 5.7–5.9, 5.14, 5.15, 5.17 and 5.20 of the fixed published original L063. It supplies the common argument for Lemmas 12, 17–19, 24, 25 and 27 of the retained appendix L064, with the published parameter restrictions stated explicitly. The retained appendix begins at Lemma 12; it does not contain the advertised proof of published Lemma 5.1.

Theorem 2 and Theorem 3 replace the simple and projective portions of the unpublished classification cited as Proposition 4.6. They may be used in the separate generic subfield argument. The cyclic, dihedral and affine arguments explain their solvable counterparts, but no claim is made here that every literal assertion about uniqueness or class fusion in that proposition is correct. In particular a representation of an abstract subgroup and an embedding of that subgroup in a larger group are different counting problems.

The small bounds are finite integer calculations, not a group census. The classical subgroup classifications, their field restrictions, the stated exceptional isomorphisms and fusion facts remain attributed prior results. This manuscript supplies the common elimination and the elementary two-transitivity arguments. It does not establish all configuration counts, complete either original, solve an open conjecture, or claim Lean verification or a compression factor.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, On the rank two geometries of the groups PSL$(2, q)$: part II, *Ars Mathematica Contemporanea* **6** (2013), 365–388. The separate retained appendix has 21 physical pages and ends at Proposition 30.

[2] L. E. Dickson, *Linear Groups: With an Exposition of the Galois Field Theory*, 1901; later Dover reprint.

[3] E. H. Moore, The subgroups of the generalized finite modular group, *Decennial Publications of the University of Chicago* **9** (1904), 141–190.

[4] M. Suzuki, *Group Theory I*, Grundlehren der mathematischen Wissenschaften 247, Springer, 1982, p. 417.

[5] P. J. Cameron, G. R. Omidi and B. Tayfeh-Rezaie, 3-Designs from PGL$(2, q)$, *Electronic Journal of Combinatorics* **13** (2006), R50, doi:10.37236/1076.